

Heterogeneous Multi-Agent System for Brain-Computer Interaction in Routing and Forwarding With Memristive Neuron Networks

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Abstract—In this research, we aimed to design a novel brain-computer interaction (BCI) approach in routing and forwarding mechanisms to realize a networking multi-modal and multi-brain linked control. In recent years, the field programmable gate array (FPGA) has become a popular choice to construct a heterogeneous network for routing and forwarding processes due to its ability to complete a high performance computation in single node. Conventional BCI mode focus on capturing the dynamic EEG signals from multiple electrode channels and this concept neglects the existing complexity and diversity of connecting information between two brain regions. To this moment, we build a heterogeneous multi-agent system (HMAS) architecture to simulate the memristive neuron networks (MNN) with a set of logic units for multi-agent in the Internet. Specifically, we first feed the non-linear features into an intelligent router to adjust their routing and forwarding processes. Subsequently, at one-time step, the output of all testing nodes is written into a heterogeneous logic space, which mainly consists of the sub-agent, forward port, and memory memristive neuron. In the FPGA, each programmable cell selectively integrates and stores a huge number of motion information between multiple interacting channels. We can achieve the networking cooperative functions of forward and route through FPGA large-scale computation. According to the extensive experimental results on several data, we validate the effectiveness of the proposed HMAS-BCI architecture to compare the optimization method of the multiple simulations.

Index Terms—Heterogeneous multi-agent system, brain-computer interaction, routing and forwarding mechanism, MNN

1 INTRODUCTION

RAIN-COMPUTER interaction (BCI) is the significant control science and technology that occur in our brain and it is attracting substantial attention from researchers [1], [2], [3]. This dynamic interaction mode usually involves an idea from the direct thinking to control a machine, which is concurrently inter-related with each intelligent agent in routing and forwarding mechanisms (e.g., some devices are connecting together, some self-organization intelligent routers are handshaking with each other). In most cases of routers' fabric processing, the concurrent inter-related heterogeneous wireless networks (HWN) [4] between the multiple routing nodes and mobile nodes are strongly interacting. It has been shown that the conventional BCI model among multiple agents, rather than a single node, can contribute a

network structure and a high prediction accuracy for the future intelligent interaction [5] in BCI.

Two main types of models exist for the BCI problem in feature signals sampling and the process of routing and forwarding step. One model (e.g., [6], [7]) is magnetic resonance imaging (MRI) to generate the dynamic feature image from our brain, and then predict the activity of brain by image processing between two regions. However, this model, i.e., regarding human interactions as multiple single-person's actions, ignores some inter-related motion information and brings with some irrelevant individual motion information. The other solution (e.g., [8], [9]) is the electroencephalography (EEG) to extract a non-stationary random signals in person's brain, and then to train an interaction recognition model and process the signals for the last calculating in a traditional BCI system.

In recent years, due to the powerful ability to program heterogeneous hardware frames [10], Field Programmable Gate Array (FPGA) [11], [12] has proven to be successful for various BCI dynamic recognition in its tasks. Therefore, we consider exploring the multiple intelligent agents among a group of routing nodes by a new routing and forwarding mechanism. However, the conventional BCI is a single node independently and does not consider the concurrent inter-relation of dynamics among multiple routing nodes. A straightforward way to overcome this limitation is to below: 1) separate servers' controlling signals and EEG sampling signals at the processing of BCI (e.g., consider interacting routers as a whole); 2) utilize FPGA platform to model a single-agent and multi-agent plan respectively, and then fuse

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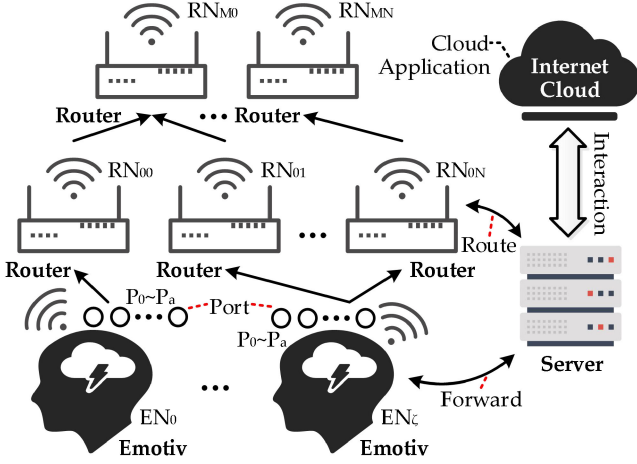


Fig. 1. The framework of the proposed Heterogeneous Multi-agent System (HMAS) for Brain-Computer Interaction (BCI) in routing and forwarding mechanism with MNN.

the single and global of Ethernet ports in high-speed route and forward computations. However, both methods neglect the inter-related dynamic features of Internet cloud interactions in multiple routing nodes.

To this end, we set up a novel Heterogeneous Multi-Agent System (HMAS) for Brain-Computer Interaction in Routing and Forwarding Mechanism (RFM) [13], [14] with Memristive Neuron Networks (MNN) [15], [16]. The system consists of the EEG signals sampling device “Emotiv”, the actuator of routing and forwarding “Router”, the net controller “Server” and the interactive cloud application “Internet Cloud”, as shown in Fig. 1. Specifically, for each Emotiv node (EN), we first feed the multiple ports into a neighbor routing node to model a dynamic forwarding network, which describes a long-term motion information in the whole HWN. Then, all inputs and outputs of these multi-agent FPGA logic units are fed into multiple brain regions for interactive interfaces, which mainly consists of FPGA logic sub-memory tool, multiple fabric forwarding ports, and multi-path routing tables. In the connection of double brain regions, we store the massive connecting information from FPGA high-speed channels. Following the HWN and logic cells allow a massive computation to enter a networking process, which selectively integrates and stores the inter-related linkage information of brain regions to reveal the real intentions of human brain. Overall, all possible linkage information at each time step is jointly modeled by a cooperative control on a routing node’s routing and forwarding processes. At the last time step, the software defined networks (SDN) scheme [17] is an Internet cloud representation of the large-scale routers’ activities. Extensive experiments on various consequences are conducted to evaluate the performance of HMAS-BCI compared with the state-of-the-art methods.

In this paper, we describe our researches in designing, implementing and testing HMAS-BCI model in routing and forwarding mechanisms. The main contributions of new research in this BCI architecture are summarized as follows:

- 1) We propose a novel and unique network structure: Heterogeneous Multi-Agent System (HMAS) architecture to build the dense connection for multi-agent

system with multiple smart terminals, it can achieve the networking brain-computer interactive functions for all agents.

- 2) We combine the Memristive Neuron Networks with a novel concurrent routing and forwarding mechanism (MNN) to aggregate the inter-related variable “ θ ” from the memristive neuron in router activity scenes over time, by adapting EEG sampling device among multiple agents to increase the BCI sampling accuracy, make its mode abundant and improve the frequency spectrum resource utilization (FSRU) of system channel in running time.
- 3) We conduct extensive experiments to demonstrate the effectiveness of HMAS-BCI architecture. Many results show that HMAS model can provide both high-quality EEG samples. Meanwhile, the BCI mode in networking could efficiently boost the real-time interactions.

Our current research model HMAS-BCI in routing and forwarding process consists of 2 components in the whole system: 1) the heterogeneous multi-agent system (HMAS); and 2) the brain-computer interactions for the routing and forwarding in memristive neuron networks. This novel architecture can transfer the conventional BCI model into the interactions between the network operating systems (NOS) [18] server and Internet cloud. Moreover, HMAS-BCI further improves the predicting accuracy of Emotiv and makes the BCI mode is not single in the future. Thus, in HMAS-BCI, we enhance the ability of BCI architecture in multi-modal perception, which can realize the network and intelligence of a human-computer interface (HCI) to improve the accuracy and real-time for interpretation in human brain’s prediction. Specifically, these routing and forwarding mechanisms in HMAS-BCI make the dynamic EEG signals of persons from the static brain features. And then, an extended HMAS-BCI model with multiple FPGA logic cells in MNN can concurrently learn an inter-related forwarding process for future research in multi-brain and multi-modal coordinated control strategies. Such as the multiple unmanned aerial vehicles (multi-UAV) case [19] in networking architecture of HMAS-BCI model.

The remainder of this paper is structured as follows. We first show the related work in Section 2 and the model of the HMAS with its main parts in Section 3. Next, we elaborate on the frame of BCI for routing and forwarding process with memristive neuron networks in Section 4. Then, we present the experimental studies in Section 5. Finally, we conclude this study and present future works in Section 6.

2 RELATED WORK

Recent studies on BCI focus on how to improve input rate and accuracy rate by designing reasonable inside resource management strategies in neural networks. In 1929s, Prof. Hans Berger described an electrical activity and uniquely exists in the human brain. In 1964s, Dr. Walter first has implemented a BCI system practically by connecting the electrodes to motion the area of a brain patient. The BCI interface can get the EEG signals and adjust parameters of algorithms in neural networks’ structure or connection. In traditional studies, we consider the more practical and

challenging scenarios for the mix of brain and computer workload as an effective interaction in EEG, and it can achieve the connecting information for the whole system to enhance the performance of BCI.

The interaction model between a human's brain and a computer machine has been studied by Santhanam *et al.* [20], they use an electrode array implanted in the monkey dorsal premotor cortex and find out a high-performance brain-computer interface for the system to improve the clinical viability of BCI in humans. More recently, Jin *et al.* [21] and Lu *et al.* [22] studied the P300 for the emotions cognition which realizes to completing the EEG signals' transmission through the P300 BCI framework. For the input rate demands of a novel brain-text communication that serves the handwriting workload, an online brain fast input mechanism has been proposed by Willett *et al.* [1]. Our work is modified to the above research literature in the following aspects: (1) the above studies construct that the platform of BCI, which is not always stable in EEG signal sampling. In contrast, in our practical actions, the unified intelligent architecture of heterogeneous networks is extracted through a routing and forwarding mechanism. (2) Rather than only consider interactive or delay-tolerant in the network of BCI, we consider the more possibility of low accuracy, a single service mode, and a low channel spectrum resource utilization of a working system when the multiple agents are in a dense connection.

To curb the escalating various problems, a large body of recent studies has been devoted to raising the recognition accuracy and BCI's channel spectrum resource utilization. To this end, the HWN [23] model and multi-agent system (MAS) [24] can be applied to construct the networks and intelligence of a BCI platform in EEG signals processing and load balancing. The intelligent routing nodes can be adapted to exploit the forwarding and routing of multiple ports and paths at different data-centers in a management [25]. While for fundamentals and development researches of HWN in edge devices, Chang *et al.* [26] consider that a node of edge computing systems should fully exploit the linkages between the artificial intelligence (AI) [27] and Internet of things (IOT) [28] while strongly processing the edge-computing-powered artificial intelligence of things (AIOT) [26]. It was succeed in networking infrastructures' resource management of prototypes to support a variety of network technologies. In 2017, J. Macker and I. Taylor [29] apply the network edge workflow tool (NEWT) to the management of some decentralized causal workflows in dynamic HWN. This method is significant for multiple BCI terminal nodes to set up a huge dense connection in HMAS-BCI architecture, but the required performance is not get from this model. For the demand of improving the sampling channel spectrum resource of utilization, many researchers devoted to studying a MAS for HWN to build interaction between a human brain and machine. In 2018, Y. Yuan *et al.* [30] establish the social network formation model which uses the representation learning methods for heterogeneous agents to retain the interpretability, we maintain the inter-agent micro-structure characteristics of most agent-based models and the macro-level structures that is focus on sociology. Sun *et al.* [31] find distributed optimization algorithms for a HMAS with a random and linked communication graph. Chen *et al.* [32]

addresses synchronization and consensus problems in networks of the linear fractional-order multi-agent systems (LFOMAS) via sampled-data control. These two cases introduces the feature of the multi-agent model and prove the two technologies can make EEG device get the network of BCI mode, however, the practical function realization can not mentioned in designing of logic units. In 2020, Wang *et al.* [33] construct a cooperative tracking approach by a MAS, they use the signals of opportunity to a distributed hybrid BP-based algorithm for cooperative tracking and network synchronization via joint likelihood consensus in multi-agent networks. The multi-modal problem is alleviated by jointing the delay and angle measurements. Liu *et al.* [34] have studied the mobile edge computing (MEC) method in networks for a distributed task migration optimization, they extended a multi-agent deep reinforcement learning method to create a system model that can make the agent and outside environment self-adaption. It is a key scheme for BCI to construct a balance between single-agent and global multi-agent.

In the above BCI structure for networks' construction, the routing and forwarding mechanism [35], [36], [37] between the single-agent and global multi-agent. In this paper, the novel networks architecture is the fundamental facility in routing and forwarding process for BCI. In 2017, Hu and Wang *et al.* [38] research a mimic NOS: MNOS for SDN approach, it enables the network more flexible, open and programmable for multiple origins and nodes cooperative control in a unified framework. Single and global agents can employ it for multi-brain and multi-origin planning. Chuang, Yu and Pang. [39] consider the SDN technology can improve the performance of data center networks. For further information on both flow-aware routing and forwarding for SDN scalability, the wireless data center is important for BCI interface to set up a novel networking architecture in EEG signals sampling. H. Zhang and Z. Li [40], [41] put forward respectively to the FRRF algorithm and PASR scheme, it have an efficiency in routing nodes during the future BCI processing. We strongly encourage these routing and forwarding methods to largely adjust the HMAS-BCI here to avoid the problems with single mode or low predicting accuracy. For more requirements of routing and forwarding process in HMAS-BCI, in 2020, P. Amaral *et al.* [42] address the challenges of a stable routing solution not covering a distributed hop-by-hop forwarding on multiple paths, which can not make ports access to EEG sampling information. They construct the networks architecture for the next step in BCI multiple ports forward and searching for brain-control nodes route path, thus achieving a good performance in BCI.

This work aims to build an efficient HMAS-BCI model in routing and forwarding process. The proposed solution for the networking of BCI can be used in a memristive synapse. In 2015, Chu *et al.* [43] propose a neuromorphic hardware system for visual pattern recognition, in this study, he make the memristor array and CMOS neuron to build a system for the first time. Hu *et al.* [44] realize the associative memory framework through a re-configurable memristive Hopfield neural network. The neural network in system as a part of adjustment for tuning the resistance of some memristors, and the prestored memory patterns can be retrieved

TABLE 1
The System Parameters of HWN

Name	Description	Value
C	The BCI device ports's number	32
N	The number of BCI terminals	10,20,50
PSEND	Power of system sending	50mW
PRECE	Power of system receiving	100mW
PMODE1	Power of 2.5Ghz system running	102mW
PMODE2	Power of 5.0Ghz system running	215mW
SCOVER	The coverage area of HetNet	200m ²

Note: # Model1 & Model2 indicates the different working frequency 2.5Ghz & 5.0Ghz in HMAS-BCI with heterogeneous wireless networks (Routers).

in the routing table directly. For example, the HMAS-BCI architecture can use this memory mode to study the routing and forwarding mechanism. When the BCI's EEG signals sampling ports get the information, the HMAS-BCI model construct the heterogeneous networks architecture, and the MNN adjust the multiple agent with the variable of EEG sampling signals's ports to describe a suitable path for routing table in HMAS-BCI system.

However, in above contributions use a heterogeneous, multi-agent and memristive neuron networks method to establish the BCI model which are an intensive interaction mode for real-time resource estimations in routing and forwarding. Our work reported in this paper uses a novel BCI model based HMAS method to efficiently identify the best brain-to-computer windows and train the intelligent routing nodes for estimating future networking spectrum resource utilization in an adjustment of BCI port and path.

3 HETEROGENEOUS MULTI-AGENT SYSTEM

In this section, we consider a HWN and MAS architecture in routers for BCI network composed of both continuous-time static and discrete-time dynamic. The heterogeneous multi-agent system (HMAS) model was designed, and its order to processing algorithm was built simultaneously. Then, we will introduce Section 3 for HMAS model.

3.1 The HWN in Routing Nodes

Similar to the traditional routing devices, a novel router architecture also suffers from a wide range of spectrum resources during an information interaction, such as edge computing, affective computing in emotions and multi-modal [45]–[47]. These variation effects lead to the design of new intelligent routing nodes that need the HWN model for routing path capturing.

For the HWN in a novel intelligent routing node. The designing structure of routing nodes is significance in the realization of networking, when dense connection for BCI terminals have worked in a networking, multi-input and multi-output (MIMO) technique makes the ports find the appropriate and low-energy forwarding network. We use the system parameters of heterogeneous networks to train the HMAS-BCI model, and it was shown in Table 1. The system is represented as (C, N, P_{SEND}, P_{RECE}, P_{MODE1}, P_{MODE2}, S_{COVER}), for the HWN construction, we choose the EPOC Flex as the BCI sample device. From 2.5Ghz to 5.0Ghz, we set up the system parameters to carefully review the basic

setup here to avoid some networking cooperative control problems with incorrect values or poorly initial settings.

In the dense connection of multiple routing nodes, the WIFI-based HWN have some particular features in HMAS-BCI model. In this paper, we propose the power and coverage area of this networking architecture and set up some initial states for the next step. The initialization parameters in the system will be formatted as a basic value in the EPOC Flex and HMAS networking. The target of the heterogeneous network in routing nodes is to make the BCI approach into multiple nodes controlling system, in this model, including the Emotiv data layers and server control data layers. We use the parameters of a router as different data transmissions in HMAS coverage area: S_{COVER}, the start value is about 200m² for simulation.

Fig. 2 is a detailed architecture of the HWN architecture in intelligent routing nodes. We use the Emotiv EPOC Flex as the EEG signals sampling device to carefully receive the usable data here and send it to FPGA process unit (ZYNQ-7000) with MNN and logic structure.

In this sub-section, we explain the overall computation process of our proposed heterogeneous networks model is visualized above. First, the transactions are read one by one and we always put a multi-agent system into the heterogeneous networking interaction between brain and computer in their interfaces. Secondly, each routing node (RN) with resource-limited and the process offloads its task to forward ports for networking execution. When we assume that each HMAS model is able to process multiple tasks simultaneously by computing processor. Specifically, T_{HF} and T_{FH} denote the maximum computation capability (i.e., MAS cycles are per second) of HAMS node N and number of tasks executed on MAS node at time slot t . Last but not least, the processing time of task J_n executed on MEC node [48] m at time slot t can be estimated as Eq. (1), where denotes how many remaining CPU cycles [49] are required for task J_n at the beginning of time slot t , ser is the serving MEC node of mobile user n at time slot t .

$$P_{TOTAL} = E_{ENV} + \frac{P_{MODE}}{S_{COVER}} \times N \times (P_{SEND} + P_{RECE}) + P_{MNN} \quad (1)$$

In this case, the power is ordered by a BCI device and MNN structure, and the multiple routing nodes extracted paths are ordered by the P_{SEND} and P_{RECE}. After rebooting this process for all paths and ports in the network system, we need to confirm a model of those routers: MODEL1, 2.5Ghz; MODEL2, 5.0Ghz. Through the minimum process of HMAS-BCI with multi-agent system, a huge number of port $\{P_0, P_1, \dots, P_N\}$ connecting pattern can be discovered in dense connections, as shown in Eqs. (2) and (3), is the two types of working frequency in a single agent (C = 16).

$$B_{MODEL1} = \iint_C (f_{IN} + f_{OUT}) \cdot f_{CORE}^2 \cdot dI, f_1 = 2.5Ghz \quad (2)$$

$$B_{MODEL2} = \iint_C (f_{IN} + f_{OUT}) \cdot f_{CORE}^2 \cdot dJ, f_2 = 5.0Ghz \quad (3)$$

Here we consider a global multi-agent of our brain and computer interface optimization problem, in which there are N routing ports $RP_1, RP_2, \dots, RP_N$ with the input and

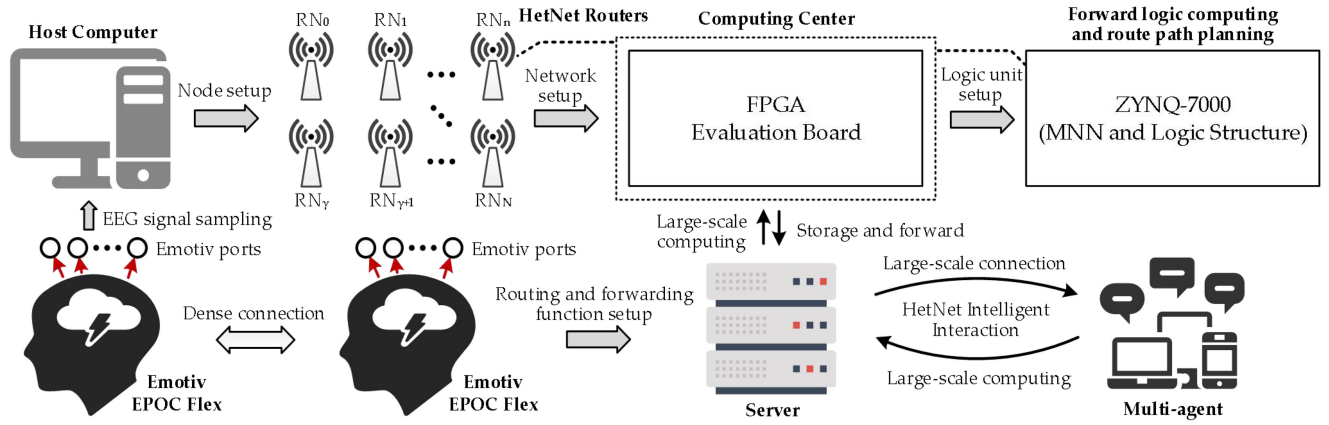


Fig. 2. The architecture of heterogeneous networks in intelligent routing nodes (Emotiv EPOC Flex for EEG signals sampling and Zynq-7000 as the core of a FPGA evaluation board), the management of networking routers is the host computer and networks' server.

output power: E_{ENV} . The multiple agents move across a huge number of MAS paths. With different migration schemes for each routing path, each frequency of a single agent may be executed on serving MAS node or migrated to connected networking for execution. In the matrix of routing table paths: T_{PATH} , it can be figured out from Eq. (4). We assume that the B_{MODEL} and G_{RN} are mentions in this text actually exist. B_{MODEL} has two types of model which should be taken in 2.5Ghz or 5.0Ghz working order, as this is how they will be placed in your paper. For example, We set up 5 paths as experiment settings to HMAS model in the 5 throughput of BCI link dispatching times. As shown in Eq. (4), T_{PATH} is the path links in networking.

$$T_{PATH} = G_{RN} \cdot P_I^J \cdot B_{MODEL} \cdot \sum_{I=0, J=0}^{I=16, J=10} \frac{H_{HMAS} + H_{MNN}}{f_{STATE}} \quad (4)$$

In addition, the routing path similarity loss is also used in our structure. In other words, an additional routing classifier G_{RN} is trained to whether the f_{STATE} features from HMAS model matrix H_{HMAS} or MNN model matrix H_{MNN} , while P is trained to fool B_{MODEL} . The domain classifiers 2.5/5.0Ghz and the feature extractor f_{STATE} are expected to equilibrium. In this way, the contribution can learn from the MAS attention networks [50]. The domain adversarial similarity loss is in Eq. (3). The overall objective function of the proposed method can be expressed as M. Where ζ is a hyper-parameter. Finally, these strategies [51] tend to trigger and benefit each router during the architecture's optimization. The 2.5Ghz and 5.0Ghz should be called out in sequential order, as this is how they will be placed in your paper. For example, avoid referring to WIFI routing nodes in the first paragraph of the article unless EEG will again be referred to after the reference to Eq. (4).

To further extract effective semantics, we also apply an attention mechanism to boost the knowledge transfer by optimizing the relationships of the samples. The output of the heterogeneous networks can produce an ideal target in networking labels, which leverages simultaneously the knowledge across multiple related domains, and the label propagation turns out to be more effective with the MNN features for multi-agent system.

3.2 Multi-Agent System

Multi-agent system (MAS), the global variable in HMAS-BCI architecture from a wide range of process algorithms during the EEG signals transmission and forward, such as RF signals receiving, IF signals sampling and data stream processing. These environmental variation effects lead to the multiple agents having variable actions and rewards to create a state between agent and environment.

Fig. 3 illustrates the working structure of a multi-agent system. In this model, the agent is affected by the external environment, the environment provides the current state to the agent and agent responds an action to the environment. In this cycle, we construct some multi-task and multi-actor for routing and forwarding process. We set up the multi-actor in matrix P , and the multi-task in matrix Q . To build the relationship between the agent and environment, we set a router networking to balance the variable of P and Q . The distributed task migration [52] is to appear in agent edge computing, it should be taken as a reinforcement learning approach in following neuron networks model to adjust the routing path and forwarding port.

Finally, we verified the classification loss of different networks in multiple target domains. The results in this section indicate that the performance of MAS attention network G is better than the feature extraction networks F , which means that it is feasible to optimize the Q using P . In this time, we determine the different states and consider a good action for the environment to achieve a great performance in EEG signals sampling. The routers indicate that the

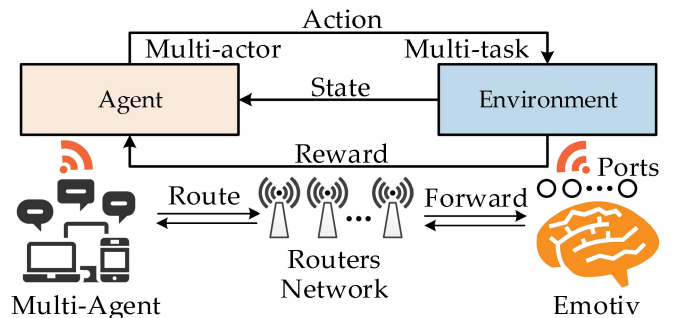


Fig. 3. The working structure of a multi-agent system (MAS).

performance of MAS attention network G is better than the feature extraction networks F , which means that it is feasible to optimize F using G .

3.3 Processing Algorithms

The above discussion is all about the multi-agent systems in the HWN model. To take the integrity of our proposed methods, we need to discuss the processing algorithms in this sub-section. Now, let us put our eyes back on Section 3.3. These variation effects lead to the intelligent network system having the functions in single-agent processing (SAP) and multi-agent global processing (MAGP), we assume the S_t as a threshold between single and global models, to estimate the change in the HMAS.

$$S_t = \iint_{n=0} (P + Q) \cdot dpdq \quad (5)$$

In this computing equation, we make the information as a dense connection tool of multiple parameters for P , Q . We integrate the sum of P , Q to review the heterogeneous networks' power here to avoid blocking problems with an incorrect order or poorly formatted system running files, a single-agent EEG signal processing algorithm in HMAS architecture: SAP is as follows.

Algorithm 1. Single-Agent Processing Algorithm

Input: C , N , $PSEND$, $PRECE$, $PMODEL$, $SCOVER$

Output: forwarding ports matrix F

```

1: Initialize the system and all parameters
2: PortFinder (single{Node},  $S = [P, Q]$ )
3: while routing is incomplete or congestion exists do
4:   SerialMaze ({ $P_i$ } { $F_j$ })
5:   update history costs of the forward ports;
6: end while
7: end PortFinder
8: for each unforwarded or congested net ports  $N_p$  do
9:   remove the value of  $RT_i$  if exist;
10:   $P_j \leftarrow \text{MazeExpand}(RT_i, t_{ij})$ ;
11:   $RT_i \leftarrow RT_i \cap \{P_j\}$ ;
12: end for
13: update history costs of the forward ports, output  $F$ ;
14: end SerialMaze
15: redesign the routing and forwarding mechanism;
16: end

```

In the process of above Algorithm 1. We get a suitable single-agent forwarding port, and know the interface of the forwarding process in routing nodes' networks. The key works were the threshold determination and we need to achieve the dynamic change in routing paths. In order to test the synchronization and real-time of our proposed method, we implement a HMAS model to use PortFinder function. SAP algorithm shows the forwarding ports that run the heterogeneous networks architecture, and multi-agent global process: MAGP, where the reproduced keys (or seeds) P are equal to the original Q from SAP. Our SAP algorithm performs the enrollment algorithm of the HMAS scheme by using the PortFinder function. And then the prototype performs the multi-factor fuzzy path planning tables reproduction with MAGP, where the recovery of key P , Q are successful.

Algorithm 2. Multi-Agent Global Processing Algorithm

Input: routing path, forwarding path, access port

Output: routing paths table G

```

1: Initialize all parameters
2: for  $Iter = 0$  to  $Iter_{max}$  do
3: Compute the fitness for all individuals  $P$ ;
4: Obtain the CDF based on the fitness values;
5: Run selection operator to get  $P$  candidates;
6: for  $i = 0$  to  $\text{length}(P)/2$  do
7: Randomly choose two individuals;
8: Generate a random number  $Pr$ ;
9: if  $Pr \leq P_c$  then
10: Run two-point crossover operator;
11: end if
12: end for
13: PathFinder (multiple{Nodes},  $S = [P, Q]$ );
14: for  $i = 0$  to  $\text{length}(P)$  do
15: Randomly choose an individual;
16: Generate a random number  $Pr$ ;
17: if  $Pr \leq P_m$  then
18: Run routing planning operator;
19: end if
20: end for
21: Update value  $P$  and  $Q$ , output the routing table  $G$ ;
22: end for
23: end

```

To correct the paths deviation from the matrix F which is FSRU of the internal and external EEG data, we utilize the EEG signals decoding operations at the multi-factor and global reproduction algorithm with MAGP. Since the limit of power depends on the P_{TOTAL} and model setting, it is significant to choose some proper parameters for setup. The parameter N is the code word length of the (n, k, t) -FSRU error correction code. For our experiments, we set n to 500 after checking the candidate parameters of routers. The parameter P is the correction threshold of the error correction. We demonstrate various test results according to proper parameters and architecture in the next section. Since PathFinder and the size of a router is 32×32 ($= 1024$), we get $Iter$ with a size $10 \sim 50$ after the forwarding process (see SAP). Therefore, in our networks for checking the HMAS model, we use all these I, J for $i = 0, \dots, 5$; $j = 0, \dots, 5$.

4 BRAIN-COMPUTER INTERACTION FOR ROUTING AND FORWARDING WITH MNN

The novel HMAS-BCI model distributes the routing and forwarding tasks to a heterogeneous networking and also marks the multiple intelligent layers after the executions of memristive neuron networks (MNN), when the neuron structure set an initial feature for the BCI, the EEG signals can be sampled in a high efficiency.

4.1 Brain-Computer Interaction

Brain-computer interaction [1] (BCI) is through the brain-computer interface [2] (BCI) to accept the EEG signals for a series of interactive application. In this section, we study the heterogeneous networking multiple ports forwarding, multiple paths describing for intelligent routing process in multi-agent system and SAP, MAGP algorithms. These

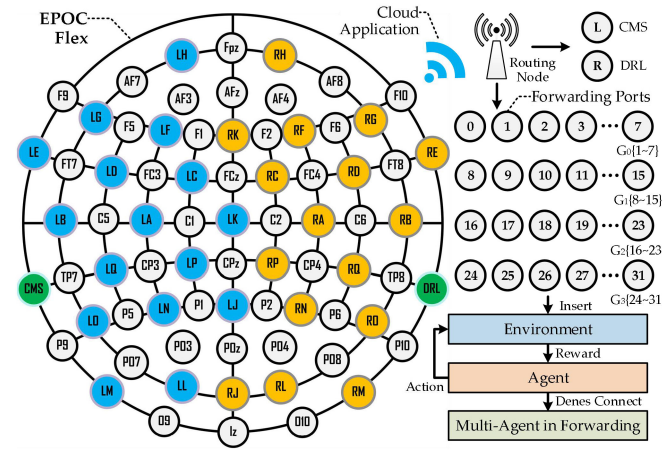


Fig. 4. The workflow of the brain-computer interaction (BCI).

approaches need an EEG signal sampling device to obtain the information, for this target, we choose the EPOC Flex [53], [54] to build this connection.

Fig. 4 shows a workflow of BCI, In this picture, we use the EPOC Flex as the system EEG sampling device. It has 32 channels (plus CMS/DRL references), each EEG sample channel is made up of some configurable on standard 72 channel international 10-20 locations. In this research, we define that the framework of EPOC Flex to forward ports. Hence, the final matrix of EEG signals in 128 SPS (1024 Hz internal) can be calculated as follows:

$$EF_{EEG} = \mu \cdot \frac{G_{ENV} \times G_{AGENT} \times P^{-1}}{(P + Q)} \times B_{MODEL}^N, \quad (6)$$

where μ represents a distinct of EEG signals sampling matrix EF_{EEG} , P , Q mean the normalized matrix value of them. Also see the forwarding ports, we consider the G_{ENV} , G_{AGENT} , as a composite matrix in environment and agents, this reward provides some helpful information and order about multi-agent action and other inserting matters. For further EEG information about typesetting mathematical equation in Eq. (6), we need to compute the state of model in EPOC Flex resolutions, the 14 bits 1 LSB = $0.51\mu V$ (16 bit ADC, 2 bits instrumental noise floor discarded).

The max slew rate is the $32.64\mu V/\text{sample}$ in EEG original sampling for the heterogeneous networks. This includes the possibility that some intelligent routing nodes will be made display routing and forwarding process to create better flow in a sub-section's study. The bandwidth is fit from 0.2~45Hz, and the high attenuation at 50/60Hz. We build in digital 5th order EEG signals filter and convent the digital channels to a FPGA logic units for HPC process in HMAS model.

In agent action and environment detection, we identify the expression where the interactions for EEG signals. We split the EPOC Flex single ADC and the coupling mode, and only the valid information with their obtained signals by routing path and forwarding ports. Then, similar as Li *et al.* [55], edit P300 is calculated between each statement in our brains and computers. One statement is considered to be transformed from the other statement, if their edit distance is less than 0.5mm. In such a way, we can locate the first buggy statement and the last buggy statement, both of

which have the edit distance lower than 0.1 with a statement in the patch. After detecting the first buggy statement and the last buggy statement, we can locate the buggy code block.

4.2 The RFM With Memristive Neuron Networks

The routing and forwarding mechanisms (RFM) with the memristive neuron networks identify and forward ports types from multiple neurons documentation. We choose API at synapse input (i.e., interfaces or interaction) rather than neural network (i.e., schemes or fields) due to choice reasons. On the one hand, the number of usage scenarios for neuron networks is very limited, since API synapses may not have complex behaviors. On the other hand, the extracted code samples for routing types usually cover the usages of forwarding.

In this sub-section, we choose routing and forwarding as case studies, since they are well established with different characteristics and attract millions of FPGA logic units. The selected routing paths and forwarding ports are organized as a set of MNN neurons in networking architecture, each of which studies a scheme in detail with a uniform formatting transmission. By parsing optimum ports and paths, synapse can correctly extract the corresponding MNN type.

As we all know, brain-computer interfaces for routing and forwarding do not contain concrete implementation in networks, thus we only extract class EEG and realize ports forwarding. It means that we do not enrich interface EEG signals with code samples and some usage scenarios. To accelerate neural networks programming and promote networking efficiency, developers tend to use non-linear EEG signals (i.e., P300) by Emotiv. Hence, we extract non-qualified neuron from MNN to exactly logic unit in FPGA platform. The same as Pande *et al.* [56] and Zhu *et al.* [57], if there exist routing path and forwarding port collisions for the same neuron in different stores, we distinguish them by using fully-qualified weights. For example, we use "Route paths" and "forward ports" to distinguish the two "Threshold" value. Similar to Chen *et al.* [58] and Liu *et al.* [59], when extracting class neurons from FPGA, we only retain the synapses inside the "ports" and "paths" to avoid novel mechanisms with different things. Finally, we obtain 2305 CLK and 1636 PLL, respectively.

As shown in Fig. 5, we introduce detailed information about the FPGA platform for BCI as a function of routing and forwarding process in an intelligent routing node. We discuss FPGA and MNN techniques applied to improve the efficiency of HWN construction with our proposed HMAS-BCI model. At first, we will describe the overall workflow of the FPGA logic memory and routing table matching integrated with a port forwarding mechanism. Second, we introduce the synapse on MNN for a server and compute the minimum load and energy consumption. Third, we present the adaptation of the SAP and MAGP algorithms over the RF front end to get a nice forwarding port and routing table. Finally, we discuss how to extend the proposed mechanism to multiple intelligent resources.

The overall working mechanism of HMAS routing and forwarding process in some intelligent routing nodes, as is implemented in our proposed HMAS model, is illustrated

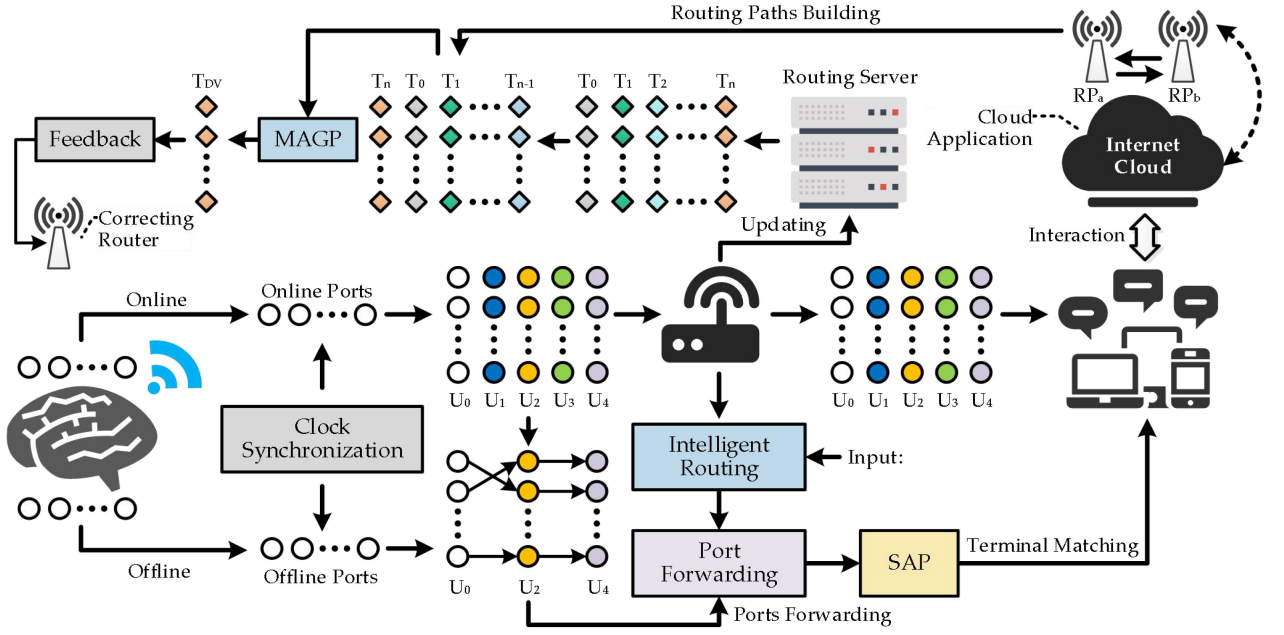


Fig. 7. The HMAS-BCI architecture in process of forwarding ports and routing paths (T means the routing table, T_{DV} means the determined value of routing table, U means forwarding ports and RN means the routing nodes).

Since the state of G_{PORT} and G_{PATH} are given in Eqs. (7) and (8), the expectation of the forward and route can be readily derived. As shown in Eq. (6), the charge to a service request is decided by parameters μ , P , Q , G_{ENV} , G_{AGENT} . In this paper, we focus on homogeneous multi-agent systems and thus do not consider its server heterogeneity in the pricing model (i.e., routing and forwarding function). However, our HMAS networking architecture can be extended to heterogeneous multi-server systems using this method. The architecture of heterogeneous multi-server systems presented in indicates that these parameters considered in our HMAS-BCI model are sufficient but need different settings to capture the server heterogeneity. For example, our feedback should allow the instruction-related parameters of a routing paths building to be changed with the EEG data and the server control data to for the HAMS. We adopt the routing nodes' Internet cloud applications which indicate the demand for services is non-linear with the $\{P_m, P_{var}, \dots, P_n\}$ (quantified by SAP and MAGP).

$$G_{PORT} = [P_m, P_{var}, P_{var+1}, \dots, P_n] \quad (7)$$

$$G_{PATH} = [RN_p, RN_{var}, RN_{var+1}, \dots, RN_q], \quad (8)$$

where P_{var+1} , RN_{var} ; $n \leq C$ are constants representing the basic ports and forwarding ports demand in routers. Note that the calculation of G_{PATH} is independent of the routing paths $\{RN_p, RN_{var}, \dots, RN_q\}$ demand function used here.

Next, we analyze the computation complexity of HMAS-BCI architecture in process of forwarding port and routing path. The overall process of HMAS consists of distributed sampling function. In this distributed sampling stage, each node pair in all equivalent classes of networking will be compared in the SAP. Thus, the worst case complexity of the EEG signals sampling process is path planning, where μ , P and Q denote the threshold, the number of ports and the number of paths. The complexity of parallel computing results are G , where SAP, MAGP denote the algorithms of

testing that will be validated. In the worst case of the whole system. Thus, the architecture of HMAS model distributed validation process is divided into online and offline.

The hardware environment at the client side consists of a PC running Windows 10 with 12 GB of RAM while the server side is implemented as the EEG signals stream in EmotivPro environment as discussed in Section 3. While we packet a session file in Matlab format, we send two formats of hybrid data: one in EEG data standard P300 format and the other in server control format. The former has low resolution and hence good for online forwarding and games as it requires less bandwidth while the plane is divided into two layers on a routing path planning. The direct CLK and BUFR formatted skeleton data helps a recorder or annotator of FPGA logic to play rich and high resolution decision-making applied to the animation and visualization. The relational database that stores routing nodes, cookie, session, DRAM, and other relational data is implemented in multi-agent and Internet cloud databases.

5 EXPERIMENT STUDIES

In this section, we will set up the HMAS-BCI and similar to the other existing BCI modes for MNN and present the experiment performance evaluation including experiment setting, management of HMAS-BCI model in routing and forwarding mechanism in MNN and simulation results.

5.1 Experiment Settings

In the sub-section, we build an experimental scenario that can simulate the HMAS-BCI model for future discussion, some EEG data stream may have to be moved from where they appeared in Emotiv. The routing and forwarding of routers will be made as an HWN to achieve the Internet of Brain (IOB) mode for future BCI applications. We will finish this novel BCI application for a networking frame with

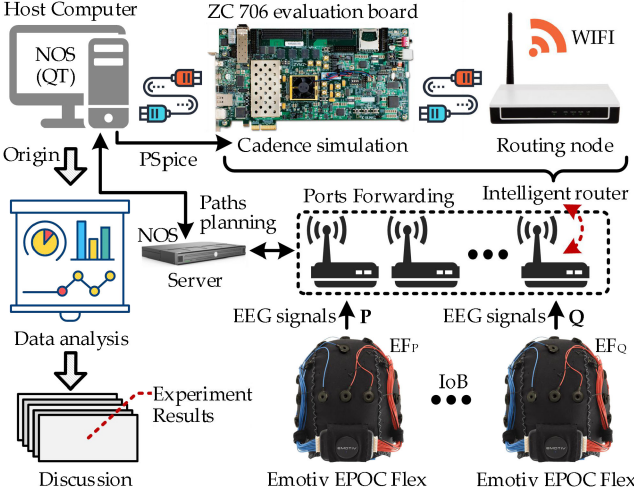


Fig. 8. The experimental scenario setting for HMAS-BCI architecture.

HMAS structure, and solve a series of problems in traditional BCI mode.

Fig. 8 exhibits the experimental scenario setting of this research for HMAS-BCI. The host computer appears as a core of this simulation. It was combined with the Lenovo 90HJS03H00 which equips with a 3.2Ghz Intel(R) core(TM) i7-8700 CPU. The operating system (OS) is Window 10 professional 64-bit. The memory has 8192Mb RAM, and the storage space is about 604Gb. We use Cadence 17.4 as the integrated development environment (IDE) of ZC706 evaluation board and AM186CC ISDN router. We set up an environment for intelligent routing nodes which based on Matlab 2021, and simulate the routing and forwarding process to obtain the results of our proposed architecture. For the EEG signals sampling, we make the Emotiv EPOC Flex as the data acquired equipment to get the matrix P and Q for the next forwarding ports. In path planning, we use the network operating system (NOS) to manufacture an application server, its control command comes from the host computer which is based on the Qt Creator IDE.

5.2 The Management of HMAS-BCI Architecture in Routing and Forwarding Mechanism With MNN

To our knowledge, the whole of research works applies management to energy efficient routing and forwarding problem in HMAS architecture for BCI. And related work mentioned is suitable for experiment comparison because the characteristics and performance of forwarding port and routing path in routers are always change. Since EEG data stream is usually unbalanced in transmission and the load balancing proportion of RP_{ports} , RP_{paths} is usually low, we use B_{MODEL} score as the performance metric which is computed in Eq. (9), the OptimPort function as a operation for RP_{ports} to get the positive relationship with G_{PORT} .

$$G_{PORT}^N \propto \text{OptimPort}\{RP_{ports}\} \cdot \frac{P_{MNN}}{\sqrt{B_{MODEL}}} \quad (9)$$

$$G_{PATH}^N \propto \frac{\text{OptimPath}\{RN_{paths}\}}{P_{MNN} \sqrt{B_{MODEL}}} \cdot G_{PORT} \quad (10)$$

In the following management of HMAS-BCI mode, the average completion of all tasks in intelligent routers is similar. As shown in Eqs. (9) and (10), we combined with the

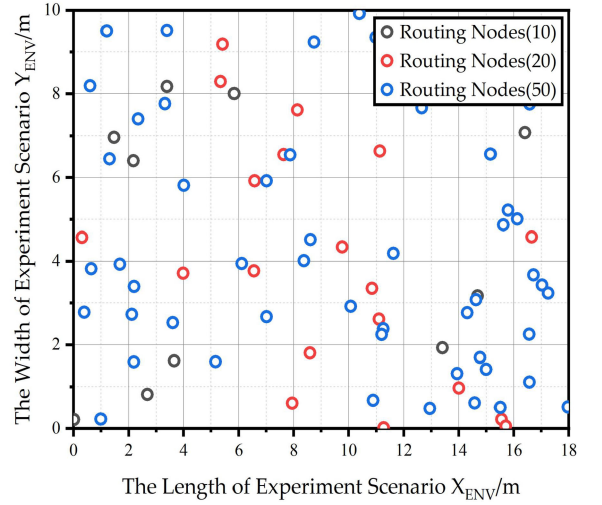


Fig. 9. The experimental scenario of routing nodes for paths planning (We set the 3 groups of routing nodes: 10, 20 and 50 routing nodes).

forwarding ports G_{PORT} and routing paths G_{PATH} for MNN with the variable P_{MNN} . For the working frequency of two modes: 2.5Ghz and 5.0Ghz, the matrix of the best path is in line with the series N . We accept this transaction before the final variations. The OptimPath function is used for this operation to get the best routing table which is in low energy consumption, and in the following research, we need to build a connection between MNN and HMAS.

Fig. 9 shows the experimental scenario of routing nodes for paths planning, we place the 3 groups of it above the picture. The X and Y axis respectively express the length and width of this simulation scenario, they are 18 and 10 meters. The number of these 3 groups are restricted of the WIFI working frequency, and they cannot realize a large area coverage. Any additional functions in a routing table will be formatted as a best-path below the low power (see Section 3.3 in this paper). In the process of routing and forwarding, as a part of the heterogeneous wireless networks, we need to compute the different features in an outdoor environment. So, the management of this study has put the HMAS for BCI and makes the MNN access to the paths routing and ports forwarding. For global path planning, we set up 10, 20 and 50 numbers routers and deploy them randomly. For example, we build these three scenarios to study the linear and non-linear features, the related researches in management of HMAS architecture, which is made up of multiple agents, we construct some FPGA platforms to get a routing and forwarding scenario, the neurons of MNN can be realized by Matlab 2021. We simulate its non-linear feature to get a variable threshold for memristive neurons and adjust the synapses structure to find out the suitable forwarding ports in routers.

In the process of the simulation experiment, we set the initial frequencies of three types of 10, 20 and 50 agents as a convergence analysis of two kinds of routing nodes $f_1 = 2.5\text{Ghz}$, and $f_2 = 5.0\text{Ghz}$. For this environment setting, we use the 2.5Ghz and its signals cannot through-wall but in the test room, the strength of WIFI signals is so good. The value of $P_{MNN} = \theta = 0.1, 0.3$ and 0.5 . The MNN variable input are $x_1 = x_2 = x_3 = x_4 = x_n = G\{\text{value}\}$, G is the input matrix of n th neuron. We set the control parameters $w_1 =$

TABLE 2
The MD and Throughput of SR in an Interaction of Routers

SR	MD (0~255)	Throughput
Connect	0	12.25Mbps
Static	1	17.50Mbps
Dynamic	5	19.25Mbps
External BGP	20	20.50Mbps
EIGRP	90	22.75Mbps
OSPF	110	24.25Mbps
ISIS	115	28.50Mbps
RIP	120	30.50Mbps
External EIGRP	170	35.50Mbps
Internal BGP	200	36.25Mbps
D2D	205	37.50Mbps
MAS	210	38.75Mbps
HMAS	220	39.50Mbps

Note: # SR(Source Routing) indicates the name of different source routing methods and MD(Managing Distances) is a distance of SR in management.

$w_2 = w_3 = w_n = 0.5$ and $\theta_1 = 0.1$, $\theta_2 = 0.3$, $\theta_3 = 0.5$. In the routing nodes, we tested the expected managing distance (MD) and its throughput of 10, 20 and 50 nodes' source routing (SR). The MD and throughput of some SR: connect, static, dynamic, border gateway protocol (BGP), open shortest path first (OSPF), enhanced interior gateway routing protocol (EIGRP) [62], routing information protocol (RIP) [63], and inter-mediate system to intermediate system (ISIS) are shown in Table 2.

In this table, we let the dead and alive nodes to describe an environment for HMAS-BCI architecture and the SR is the management of the can be especially confusing because the throughput would know whether the MD is in a good performance for SR. When we create the BCI systems, especially in complex signals sampling and processing, we should ensure that weights are thick and thin enough that when throughput of routing at thickness size, MNN still be regulate the routing paths and forwarding ports. We set all works must follow Emotiv guidelines and managing the HMAS-BCI model in routing and forwarding with MNN.

5.3 Simulation Results and Discussion

In this section, based on the HMAS model, the BCI in same initial condition and also in same experimental environment which is used in the discussion. The simulations are shown in following research. We set the special scenario in different features for HMAS-BCI architecture and computation of the task optimization in some obvious targets.

For Section 4.2, we list the route and forward features in the neurons of MNN. For example, within the memory unit, the system can cache the matrix $\{V_1, V_2, \dots, V_n\}$ in the multi-agent which followed by a close parenthesis. Based on this, experiment setting lists will be built so that the simulation results could better discuss in the next step:

1. The HMAS to effectively build a dense connection of multi-agent with multiple smart terminals, and realizing the networks among multiple agents in the brain-computer interaction, we separately test the capacity of forwarding ports and routing paths in different networks: Client-Server (CS), Browser-Server (BS), Device to Device (D2D) and HMAS;

2. The MNN model with a novel concurrent routing and forwarding process to aggregate the variable " θ " from neuron in routing nodes activity scenes over time, by adapting the sampling device of EEG signals among the multiple agents to increase the accuracy of BCI. In this simulation, we use 5000ms running time to study the change of its counts and number of dead nodes in networking architecture;
3. For this proposed of a good performance in a BCI study, we set up a throughput varies with the link dispatching times of HMAS-BCI, we build 5 paths, and in 500ms running times, we test 1000 points of each path and achieve the power point sets.

Routing, forwarding approaches and MNN computing structures, such as an causal relationship of research, and inducement, are formatted using an interactive order. In this section, the routing nodes begin with a dead and alive node, which are followed by the MD, in functions of sender and receiver, as shown in Eq. (11), it is the distance between two-type communicating routers, we use Eq. (12) to compute the total factors between sender and receiver.

$$d = \text{dist}(i, j) = \text{sqrt}(S) \quad (11)$$

$$S = \sum_{i=1, j=1}^N [\text{send}(i, j) - \text{receive}(j, i)]^2 \quad (12)$$

Where (i, j) represents the sender and receiver of routers, in these equations, d represents the distance of (i, j) and it is also equal to $\text{sqrt}(S)$. When the S is in an appropriate time, the system can computing the MD of the simulation and get a value of factor in router's sending and receiving process, in matrix G_{path} and G_{port} , when it gets a dead node, the simulation's computation can find out the EEG signals and gain the relevant parameters to testify a performance for the following works.

To further verify the uniform convergence of the multi-agent systems under heterogeneous wireless networks in proposed SAP and MAGP algorithms, five kinds of CS, BS, D2D, and HMAS networking architectures were used for simulation in experiments based on the different types of networking architecture with back-stepping technology. The results are shown in Fig. 10, in this picture, we want to get a relative superiority for HMAS architecture. Thus, the experiment setup has put forward to four networking architecture and do comparative simulation. In the Fig. a-d, we use 10 routing nodes for its communications, and in Fig. e-h, i-l, the number of nodes are 20 and 50. In this 3 scenarios, we first test the capacity of forwarding ports in average task completion time (ATCT), then we simulate the planning functions of routing paths and in this part, the experiment has build 3 scenes, they test the error rate (ER), the latency, the transmission rate (TR) of 4 different networking models.

Fig. 10a shows the forward routing paths histogram of the multi-agent systems. As seen in this figure, as a total number of forwarding ports changes, the histogram of 4 parameters roughly displays a sort of slowly decreasing first in ATCT and then HMAS networking architecture has the lowest ATCT value because it has an HWN from a different application layers, moves with a time-varying in intelligent routing nodes' forwarding ports, and tracks its desired designated ports.

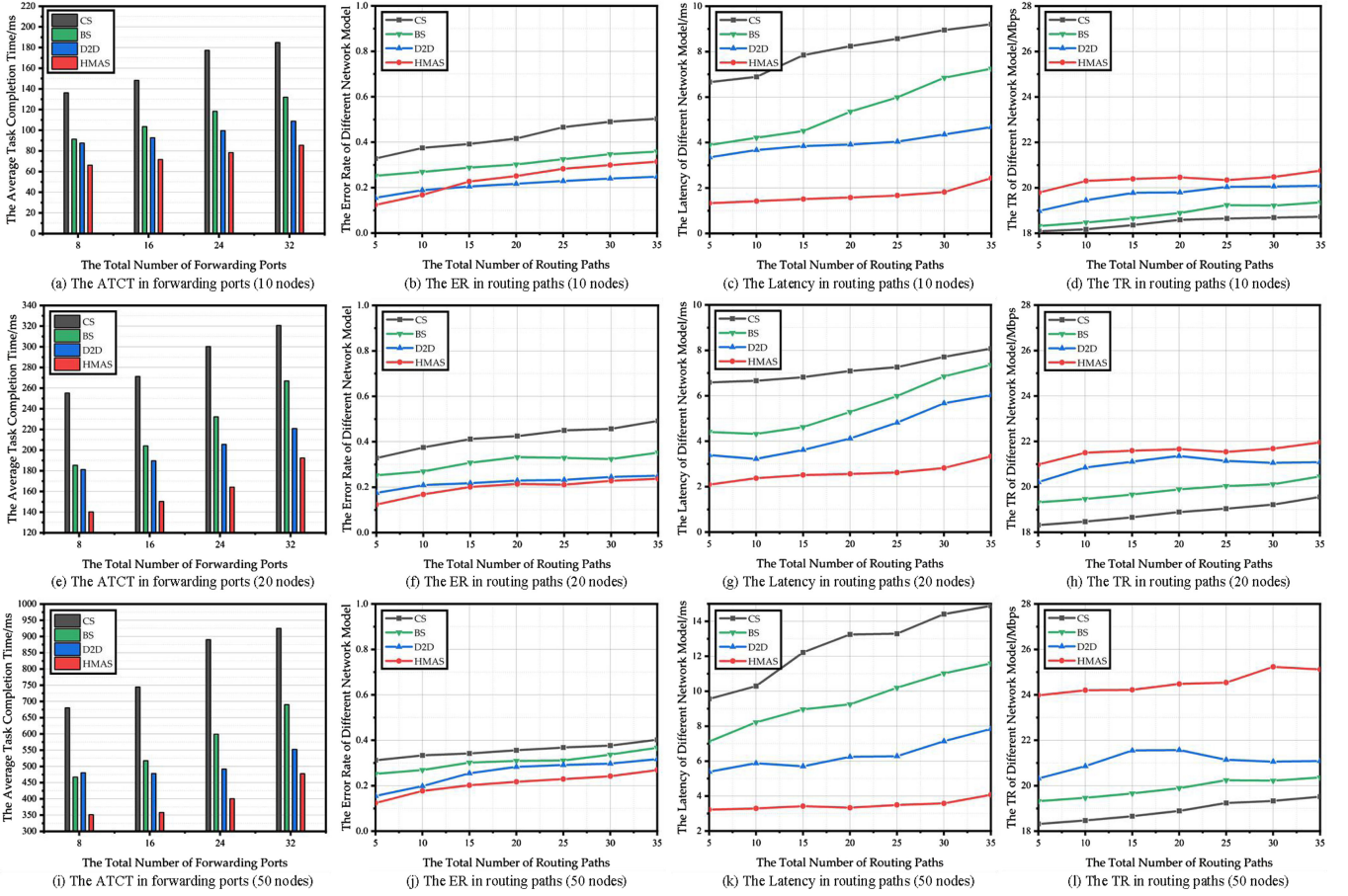


Fig. 10. The average task completion time (ATCT) varies with the total number of forwarding ports(10,20,and 50 routing nodes); and the error rate (TR), latency, transmission rate (TR) of different network model varies with the total number of routing paths(10,20,and 50 routing nodes).

Before each dynamic intelligent agent, between its EEG fabric signals ports in the process of forwarding process, when the speed is at the specified port, the ATCT between CS, BS, D2D, and HMAS show its relative superiority of proposed networking architecture. In Figs. 10e and 10i, the acceptance test 20, 50 routing nodes, the experimental results also show the relative superiority for HMAS architecture in BCI actions.

In addition, some works prepare to test the performance by examining the ER, latency and TR of different networks models. Fig. 10b shows the CS, BS, D2D and HMAS curves of ER varies with the total number of routing paths. As the number of routing paths changes, the ER curve generally presents a state of slow increase and then tends to be stable. In the process of 10 agents tracking the desired nodes, the proposed SAP, MAGP algorithms can achieve the tracking purpose quickly and effectively and cause these 10 agents to achieve the routing tables for paths planning. Figs. 10e and 10f are the 20 and 50 testing nodes, they get a similar result for ER index to illustrate a good performance of HMAS networking architecture. In Figs. 10c and 10d, they show a similar feature for HMAS to describe the latency and TR has superiority in EEG signal communications.

With the increasing number of routing paths, from 5 to 35, we use 2.5Ghz working frequency for routing nodes to communicate with each other. Figs. 10b,10f, and 10j; Figs.10c, 10g, 10k; Figs. 10d, 10h, and 10l fully illustrate the relative superiority in HMAS architecture, and it also provides proof of the proposed networking interaction mode in HMAS-BCI. It can

realize a dense connection between BCI and multiple routers, to complete the application accessing to the Internet.

Fig. 11a depicts the variable threshold for memristive neurons. The results demonstrate that a simulated neuron associated with data of paths (G_{PATH}) and ports (G_{PORT}) for routing and forwarding processes do exist. The flow data activate more for memory functions over the " θ " variable threshold from 0.11 to 0.72. While the EEG signal flows of G_{PATH} and G_{PORT} are similar to this of forward ports matrix, which both show more activation in the forwarding areas, the memristive neurons have higher stabilized responses at the threshold sites of single node. This is a memristive neuron threshold feature of single channel. In Fig. 11b, it was the multi-plane neuronal parallelism feature of MNN architecture, the energy demand has significantly higher ladder threshold responses at forward ports and channels. We use these threshold features of memory function and demand of differential energy to manipulate the hopping frequency and dead nodes in global planning of multiple routing nodes with SAP and MAGP algorithms.

Fig. 12 shows the routing process when MNN tracking routing tables. As system running time changes, the curve presents a trend of hopping from oscillation to motionless. Similarly, the formation system of MNN start to move at different thresholds and the hopping frequency is from 0 to 35. We set up the threshold " θ ", and the value are 0.1, 0.3, 0.5. Before 1500ms, since the forwarding is a changed routing path, and there is corresponding movement in the

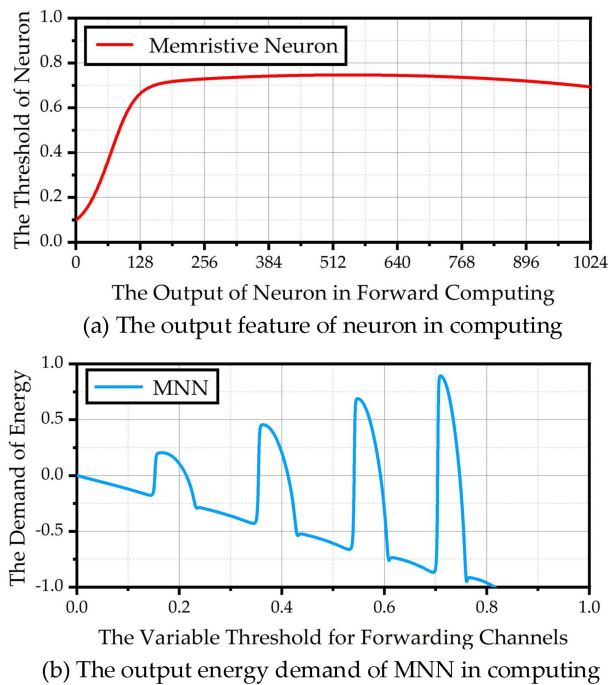


Fig.11. The output feature of neuron and the output energy demand of MNN in routing and forwarding computation.

routing nodes, the multiple agents move from different starting nodes, and the hopping frequency curve of the 3 thresholds undergo an ending because each agent has weak self-adaption ability in the routing and forwarding process, we need to regulate the routing nodes' ports and paths. After 3000ms, the multiple agents have tracked the desired ports and formed a routing table in the neurons of MNN. The relative hopping frequency of their direction y path is decreased, which can achieve a better formation in high FSRU and low power consumption, as shown in Fig. 12a, is the hopping frequency varies with running time.

Fig. 12b shows the number of dead nodes varies with running time. As running time changes from 0 to 5000ms, these 5 curves show a stepwise growth and then quickly stabilize. According to the 10, 20, 30, 40, and 50 routing nodes begin to work from the 1000ms with different test nodes. In this process, the 5 types complete the formation hopping for neurons in MNN and ending in the threshold is invariable for the weight. In 10 nodes, the neuron in its forwarding process uses the 10 steps to get the final dead nodes. Therefore, the 20, 30, 40, and 50 nodes will show a similar 20, 30, 40 and 50 steps at the end of the running time, under the control of HMAS networking architecture, we achieve a stable state in routing and forwarding with MNN, that is, the neurons in networking system exhibits a maximum FSRU and minimum power trend for BCI.

Table 3 shows the resource utilization (RU), clock rate (CR) and power of algorithms for BCI in FPGA processing of routing nodes. As illustrated in this table, in difference of the processing algorithms, RU presents the utilization efficiency for FPGA logic unit with an optimal approach. In the process of EEG signals sampling, each routing node's traces in the desired designated position are different in CR, we need to detect it to verify the good performance in path assigning of networking model. During this process, the sparse matrix

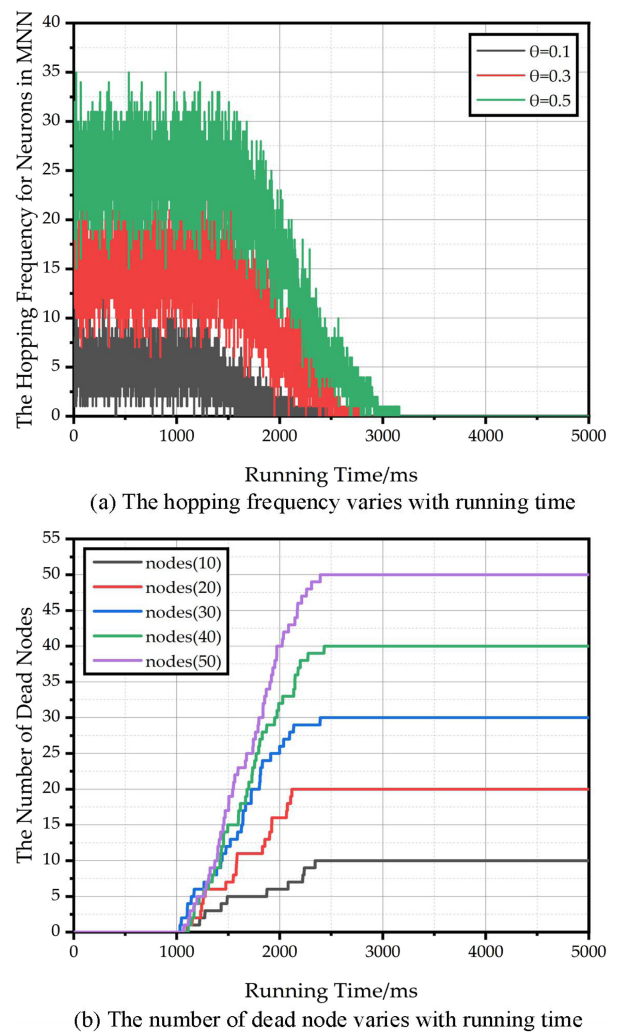


Fig. 12. The counts for MNN computing model and a number of dead nodes varies with the HMAS architecture's running time in BCI.

vector (SPMV) [64], page rank (PR) [65], single source shortest path (SSSP) [66], [67] and weighted community clustering (WCC) [68], [69] are different in CR value, the single agent is about 210Mhz, and the multiple agents are about 205Mhz.

TABLE 3
The Resource Utilization (RU), Clock Rate (CR) and Power of Algorithms for BCI in FPGA Processing of Routing Nodes

Algorithms	RU	CR	Power
SPMV(S)	8.52%	212Mhz	3.52mW
PR(S)	1.50%	211Mhz	3.55mW
SSSP(S)	5.25%	209Mhz	5.27mW
WCC(S)	8.25%	208Mhz	5.29mW
SPMV(M)	9.15%	206Mhz	5.28mW
PR(M)	15.25%	207Mhz	5.25mW
SSSP(M)	15.50%	207Mhz	6.22mW
WCC(M)	12.25%	207Mhz	6.25mW
ANN	13.25%	205Mhz	6.25mW
CNN	20.10%	202Mhz	5.55mW
MNN	22.25%	209Mhz	5.25mW
SAP	22.75%	214Mhz	5.22mW
MAGP	25.25%	215Mhz	5.29mW
SAP-MAGP	25.59%	201Mhz	5.90mW

Note: #(S) indicates the single agent of routing nodes and (M) indicates the multiple agents in the forwarding calculation with a FPGA platform.

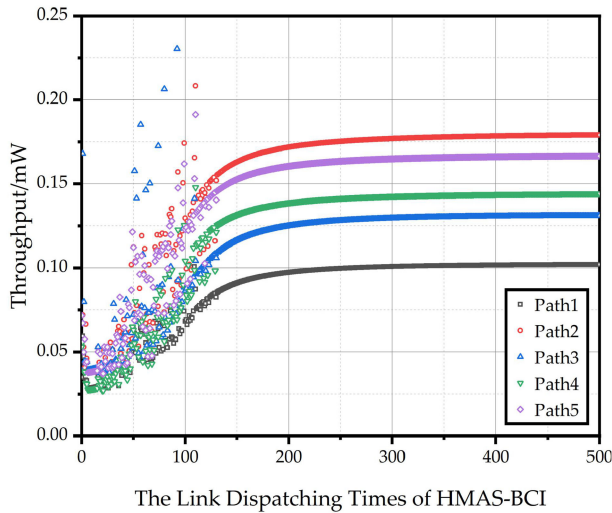


Fig. 13. The path planning throughput varies with the link dispatching times of HMAS architecture in BCI scenario.

This indicates the proposed MAS methods in routing and forwarding processes can better avoid the obstacles and in the core of power consumption for the relative distance between a pair of routers is determined to be close to each other. In neural networks, ANN is the basic style for the variation of neuron weights. Before the threshold changes, the MAGP can make the single routers in the FPGA platform for searching the adjacent routing nodes. When the threshold is similar, and the repulsive force of them is greater than its attractive force, which makes these multi-agent moves away from the original path planning. In this time, when the SAP is combined with MAGP, SAP-MAGP, which presents some optimal parameters for BCI in FPGA processing of the routing nodes. The RU is 25.59%, CR is on 201Mhz, and power is on 5.90mW. It shows that the proposed routing and forwarding algorithms of HMAS are a workable plan for BCI in networking architectures.

The experimental studies on HMAS networking are focused on a routing and forwarding process. On the one hand, HMAS-BCI can achieve up the great performance improvement on the ATCT of in forwarding process with larger scale of ports. In addition, the ER, latency, TR of 4 networking architectures increase so much with increased number of routing paths. The reason lies in that number of routing paths and forwarding ports are changed, thus the FPGA computation resource becomes more intensive. The HMAS-BCI model in neurons also has a satisfactory performance improvement in above simulations, which is compared with CS, BS and D2D networks. In Fig. 13, the path planning throughput were varied with the five-link dispatching time of HMAS in BCI scenario. Therefore, we set up 5 paths: Path1, Path2, Path3, Path4, Path5 to build a study for the linking communication of different paths.

On the other hand, we also change the system running times T_{PATHS} from 0 to 500ms with other variables remain unchanged. From Fig. 12, we find that the throughput of all tasks for Path1, Path2, Path3, Path4 and Path5 tend to be different in path planning. As we said before, in BCI for routing nodes, tasks are executed on routing node that they are first forwarding ports to until they are completed, no matter where routers that they belong is moving to G matrix. Thus

SCOVER and P_{MODE} have no effect on the power of networking architecture. For HMAS and BCI model, an optimization method of multiple agents is increased with decreased FSRU. In ports setting and paths planning, the performance improvement of HMAS-BCI distributed task scheduling grows with an acceptable range in simulation. To sum up, all experimental results that verify advantages of the proposed frameworks for BCI in systems' routing and forwarding process to construct the dense connection of multiple routers and realize an intelligent interaction.

6 CONCLUSION AND FUTURE WORKS

This work combines the heterogeneous wireless networks of routing and forwarding in a multi-model fashion to produce accurate forwarding ports and routing paths predictions for different routing nodes. The experiments show that a HMAS networking architecture better than the CS, BS and D2D in BCI with neuron networks. The current study was limited to the dense connection whereas complex deterministic models are able to cover a huge EEG signal sampling and processing (i.e., IOB). The results demonstrate that an adequately trained HMAS model outperforms other standard BCI approaches. The routing and forwarding capabilities of the network have not been investigated as they were out of the scope of this research and will be researched in future works.

Future improvements on HMAS will be focused on the training with non-linear heterogeneous models (e.g., chaos networks); the extensions of the forwarding window (e.g., 32 multiple forwarding ports); the training of the network with multiple parameters affecting the neurons in neural networks (e.g., FPGA ZC706 evaluation board in Cadence 17.4 IDE and MNN at a structure of memristive neurons); and the integration of SAP, MAGP forecasting algorithms into the Heterogeneous Multi-Agent System (HMAS) and Brain-Computer Interaction (BCI) architectures [1], [2], [3] to expand the dense connections for BCI terminals, increase the functionalities for FSRU currently offered by the MAS towards HPC and the capabilities in networks' powers.

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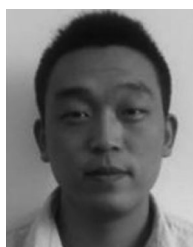
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